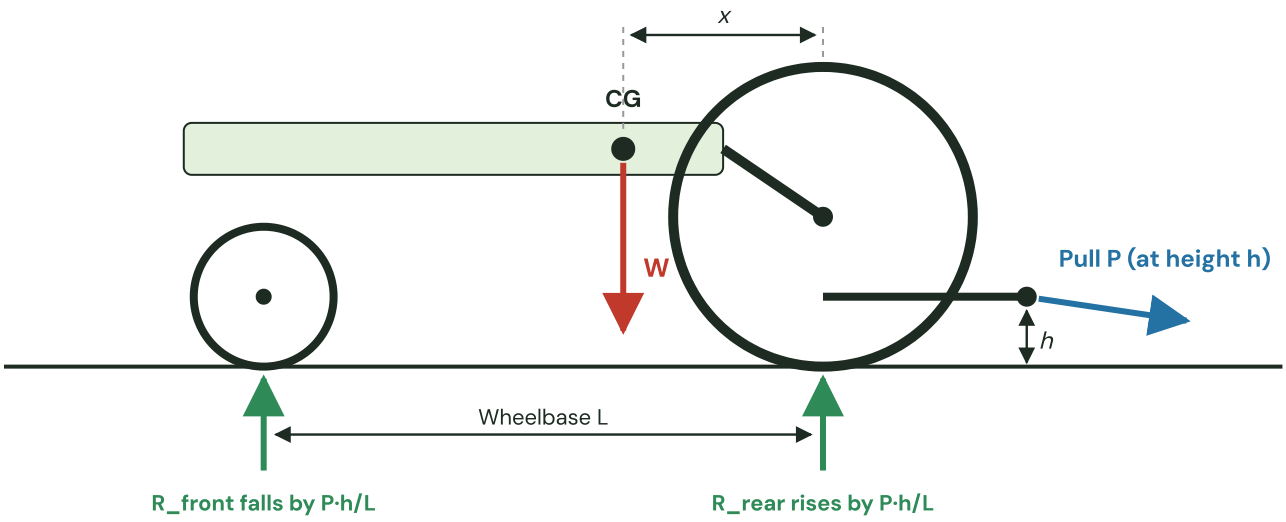
UNIT 4 · TOPIC 9(A)

Chassis Mechanics: Weight Distribution and Weight Transfer

Syllabus topic: Tractor chassis mechanics and hitching systems

Static: R_front = W·x/L (x = CG ahead of rear axle); 2WD ≈ 1/3 front, 2/3 rear

Pulling: ΔR = P·h/L shifts load front → rear; if R_front → 0 the tractor rears up (high hitch = danger)



Static reactions and weight transfer under pull (redrawn — the source figure showed the pull pointing forward)

Static weight distribution

Treat the tractor as a beam on two axles; x = horizontal distance of the CG ahead of the REAR axle, L = wheelbase.

$$R_{front} = W \cdot x/L;\ R_{rear} = W \cdot (L - x)/L$$

- A 2WD tractor's CG lies well aft: ~**1/3** of the weight on the front, ~**2/3** on the rear (4WD nearer 40/60).
- The CG is about **0.5–0.8 m** ahead of the rear axle, and low for stability.

Ex.1 W = 20 kN, L = 2.0 m, x = 0.8 m → R_rear = 20 × (2 - 0.8)/2 = **12 kN (60%)**; R_front = **8 kN (40%)**.

Weight transfer under pull

$$\Delta R = P \cdot h/L \text{ (drawbar pull P at height h moves load from the front axle to the rear)}$$

- Useful — it adds traction — but dangerous in excess: when R_front → 0 the front lifts.
- Combined with rear-axle TORQUE REACTION (the chassis tends to rotate backwards about the rear axle when the wheels lug or are trapped), this causes **REARWARD OVERTURNING** — the classic fatal accident with high hitching or trapped wheels.

Prevention and response:

1. Hitch ONLY at the drawbar (low).

2. Add front ballast.
3. If the front starts to lift, disengage the clutch instantly.

> **Ex.2** P = 6 kN at h = 0.4 m, L = 2 m → ΔR = 6 × 0.4/2 = **1.2 kN** → R_front = 6.8 kN, R_rear = 13.2 kN. Keep at least ~**20%** of the tractor weight on the front axle for steering control.

Lateral stability (recap)

$$\text{Side-slope overturn when } \tan\theta = (B/2)/h_{cg}$$

A wider track and lower CG are safer (Unit 2, Topic 4). ROPS + seat belt protect if an overturn occurs (Unit 5).

Practice MCQs

Q141. For CG at distance x ahead of the rear axle and wheelbase L, front-axle reaction is:

(a) W·L/x (b) W·x/L
(c) W(L-x)/L (d) W/2

Q142. W = 20 kN, L = 2 m, x = 0.8 m → rear axle carries:

(a) 8 kN (b) 10 kN
(c) 12 kN (d) 16 kN

Q143. A 2WD tractor's static weight split front:rear is about:

(a) 50:50 (b) 65:35
(c) 35:65 (d) 20:80

Q144. Weight transferred rearward by pull P at drawbar height h is:

(a) P·L/h (b) P·h/L
(c) P/(hL) (d) W·h/L

Q145. P = 6 kN at h = 0.4 m, L = 2 m → weight transfer =

(a) 0.6 (b) 1.2
(c) 2.4 (d) 4.8 kN

Q146. Rearward overturning of a tractor is promoted by:

(a) Low drawbar hitching (b) Hitching high above the drawbar and rear-axle torque reaction
(c) Front ballast (d) Wide track

Q147. For steering control, the front axle should keep at least about:

(a) 5% (b) 20%
(c) 50% (d) 70% of tractor weight

Q148. Weight transfer during pulling is:

(a) Entirely harmful (b) Useful for rear traction but dangerous in excess
(c) Irrelevant (d) Zero at high pull

Q175. If the front of the tractor begins to lift while pulling, the operator must FIRST:

(a) Accelerate (b) Disengage the clutch immediately
(c) Raise the implement (d) Turn sharply

Answers

Q141 — (b) W·x/L

Q142 — (c) 12 kN — *R_rear = W(L − x)/L = 20 × 1.2/2 = 12 kN.*

Q143 — (c) 35:65

Q144 — (b) P·h/L

Q145 — (b) 1.2 — *ΔR = 6 × 0.4/2 = 1.2 kN.*

Q146 — (b) Hitching high above the drawbar and rear-axle torque reaction — *A pull line above the rear-axle centre (or wheels trapped in mud) creates a moment that rotates the chassis about the rear axle.*

Q147 — (b) 20%

Q148 — (b) Useful for rear traction but dangerous in excess

Q175 — (b) Disengage the clutch immediately — *The clutch is the instant remedy — raising the implement takes longer.*

Sources: Liljedahl et al. — Tractors and their Power Units; Kepner et al. (hitching); IS 4468/ISO 730; Jagdishwar Sahay; NPTEL lectures.
